Lecture Homework Week 06 - Thursday Template

Student Information

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Student Responses

Answer the following questions

Question 1

When running the Keyboard Demo in QEMU after introducing the bug, briefly describe how it behaves differently than before. Try a few keystrokes in the QEMU window and describe what you see:

The system behaves inconsistently. Most keystrokes no longer create new lines, instead overriding the previous keystroke. Some do create new lines. Some cause letters and glitches to occur in seemingly random places on the display.

Question 2

What do you think is happening? The program continues to execute, but what happens after it returns from `irq_handler`? What C subroutine gets called over and over?

When the program returns from irq_handler it returns only a portion of the previous registers back to their original location. The other registers are left and our stack pointer no longer points to the correct location. Since our program counter is also restored from the stack, it gets loaded with the wrong value. Since the system just booted, most registers contain 0. When 0 is loaded into the program counter we execute the reset handler which ultimately calls Main() over and over.

Question 3

Look at the terminal where the buggy version is running, what messages are being printed out about the timers? Do the timers work in this buggy version?

The text is “Timer with delta zero, disabling.”
The timers do not appear to work by the message; it seems the timer(s) have been set to trigger every 0 seconds. The system has apparently detected and not allowed this for obvious reasons. The timer issue likely occurs for the same reason that we end up calling the reset_handler.